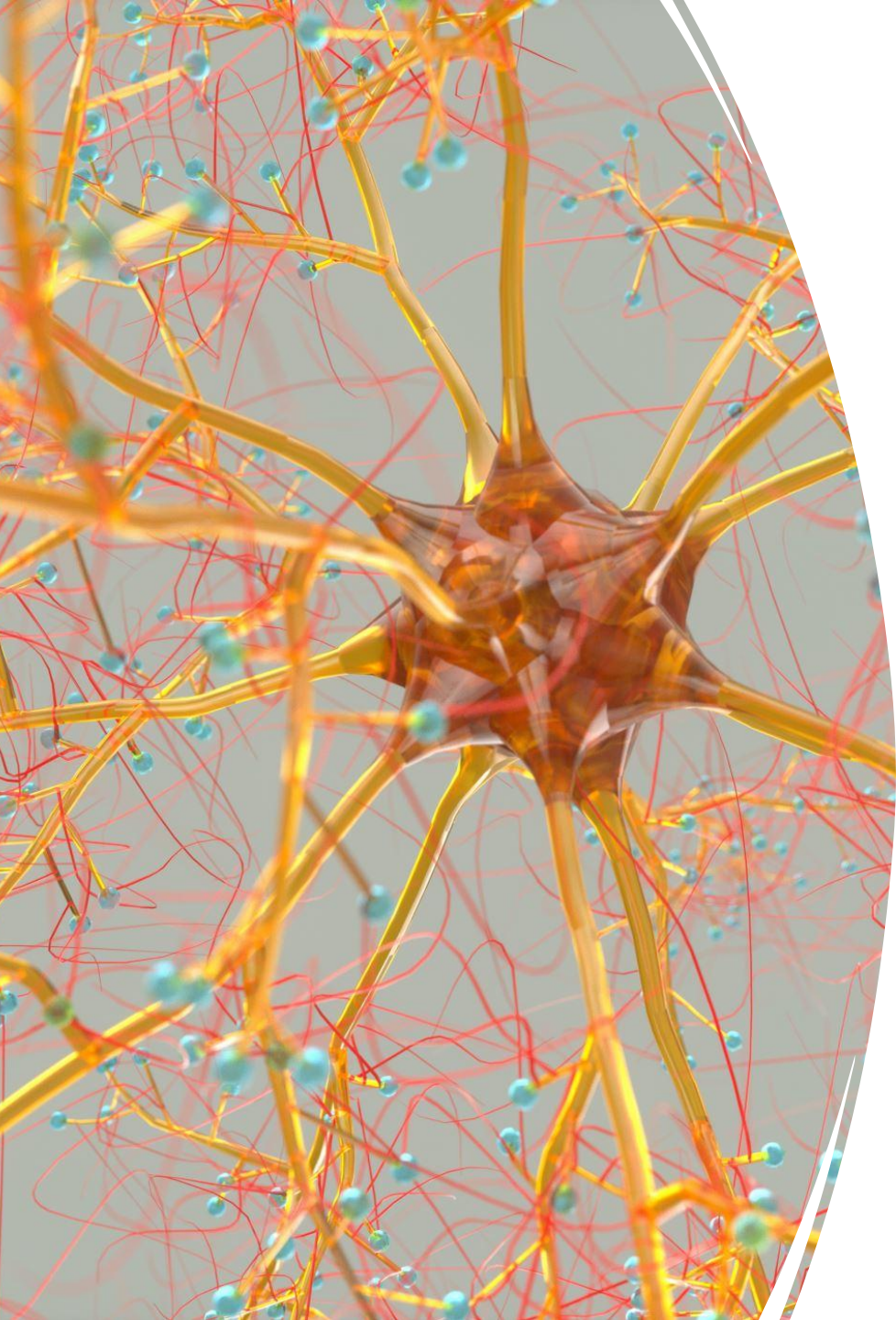




Collagen-Vascular Diseases

Addis Ababa University
Clinical Pharmacy

Definition

Collagen-vascular diseases (CVDs) are a group of chronic autoimmune disorders characterized by immune-mediated inflammation and damage to connective tissues, and



They primarily affect:

Skin

Joints

Muscles

Blood vessels

Internal
organs
(kidney, lung,
heart, CNS)

Pathogenesis (Common Mechanisms)

Loss of immune
tolerance

Autoantibody
production

Immune complex
deposition

Complement
activation

Cytokine and
interferon
pathway activation

Vasculitis and
fibrosis

Key concept: These diseases share overlapping immunologic mechanisms but differ in dominant organ involvement.

Classification of Collagen-Vascular Diseases

- **Major Disorders**
 - Systemic Lupus Erythematosus (SLE)*
 - Rheumatoid Arthritis (RA)*
 - Systemic Sclerosis (Scleroderma)
 - Polymyositis / Dermatomyositis
 - Mixed Connective Tissue Disease (MCTD)
 - Sjögren's Syndrome
 - Vasculitides (often included)

Management

- Immunosuppressants
- Biologics
- Organ-specific treatment



Systemic Lupus Erythematosus (SLE)

- Multisystem autoimmune disease
- Common in young women
- Autoantibodies: ANA, anti-dsDNA, anti-Sm
- Organ involvement: skin, joints, kidney, CNS, blood

Key drugs: Hydroxychloroquine, corticosteroids, azathioprine, mycophenolate, biologics

Rheumatoid Arthritis (RA)

- Chronic inflammatory polyarthritis
- Symmetric joint involvement
- Extra-articular manifestations: lung, heart, eye
- **Autoantibodies:**
 - Rheumatoid factor (RF),
 - anti-CCP (anti-cyclic citrullinated peptide antibodies)

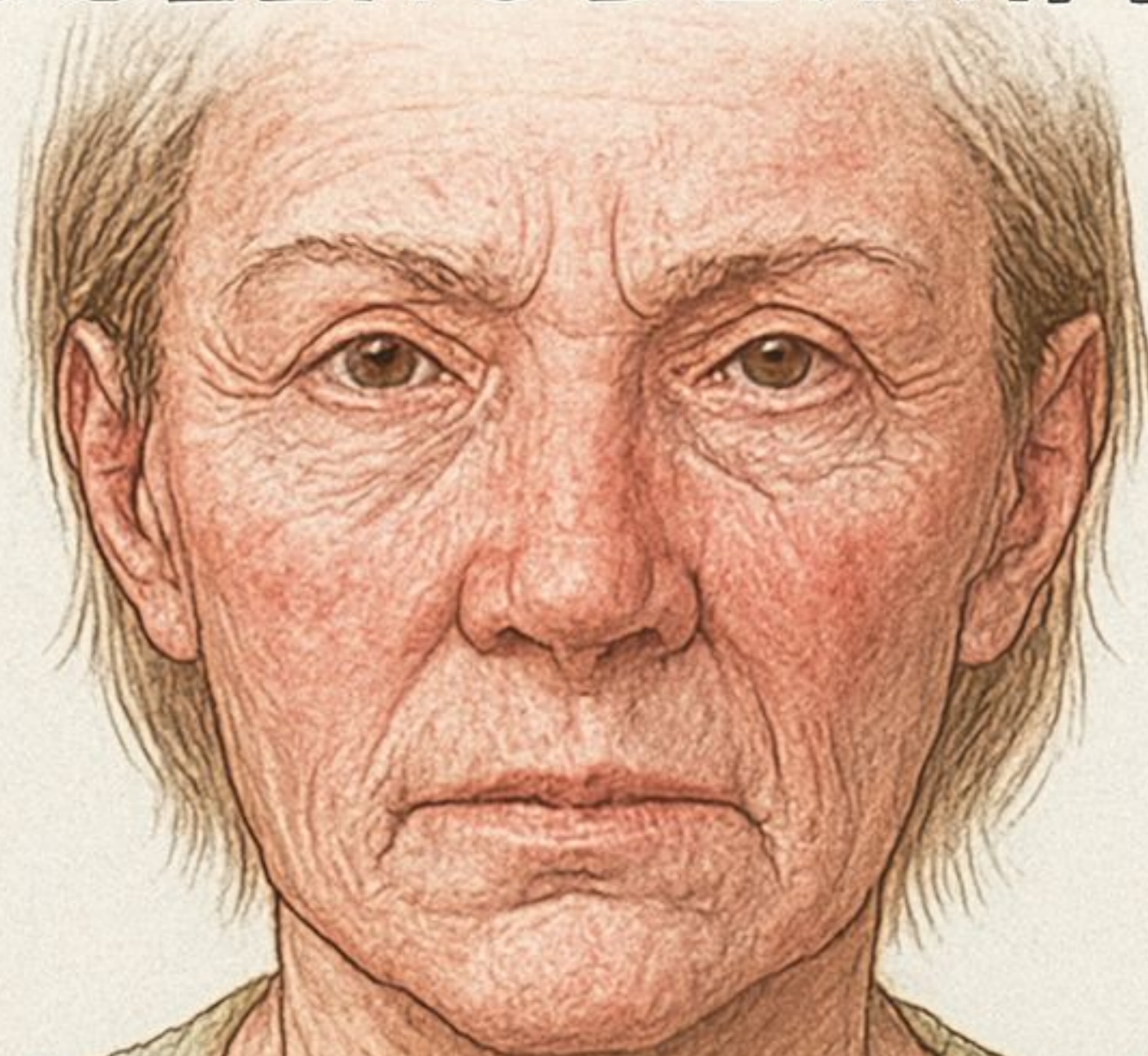
Key drugs: Methotrexate, biologics (TNF inhibitors), Janus Kinase (JAK) Inhibitors

Systemic Sclerosis (Scleroderma)

- Excessive collagen deposition → fibrosis
- Skin thickening and organ fibrosis
- **Types**
 - Limited cutaneous (CREST)
 - Diffuse cutaneous

- **Key features:**
 - Raynaud phenomenon
 - Interstitial lung disease
 - Pulmonary hypertension
 - Renal crisis
- **Autoantibodies:**
Anti-centromere, anti-Scl-70 (anti-topoisomerase I)

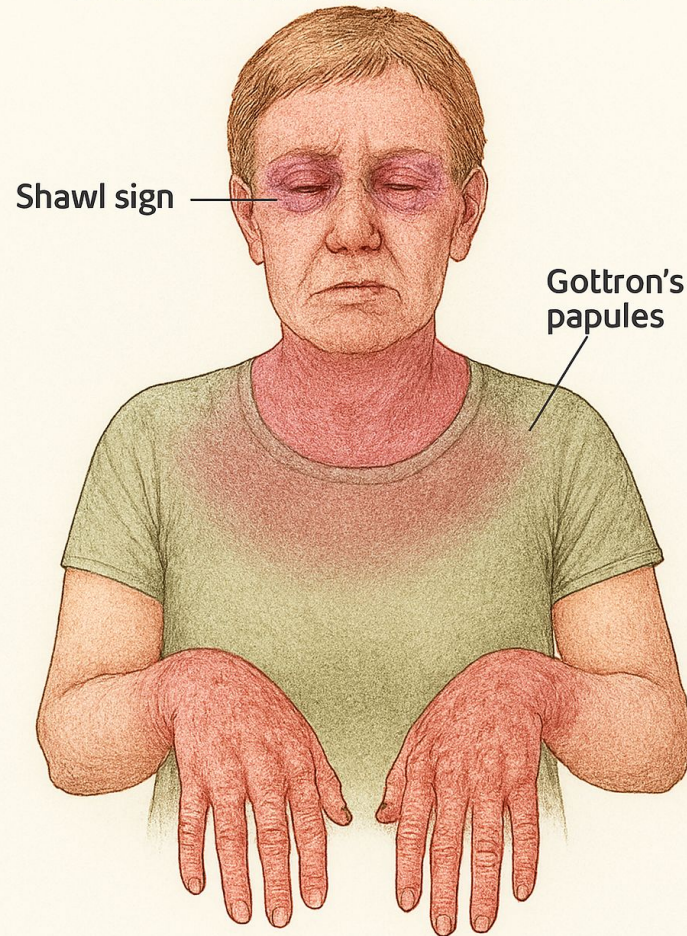
(SCLERODERMA)



Polymyositis and Dermatomyositis

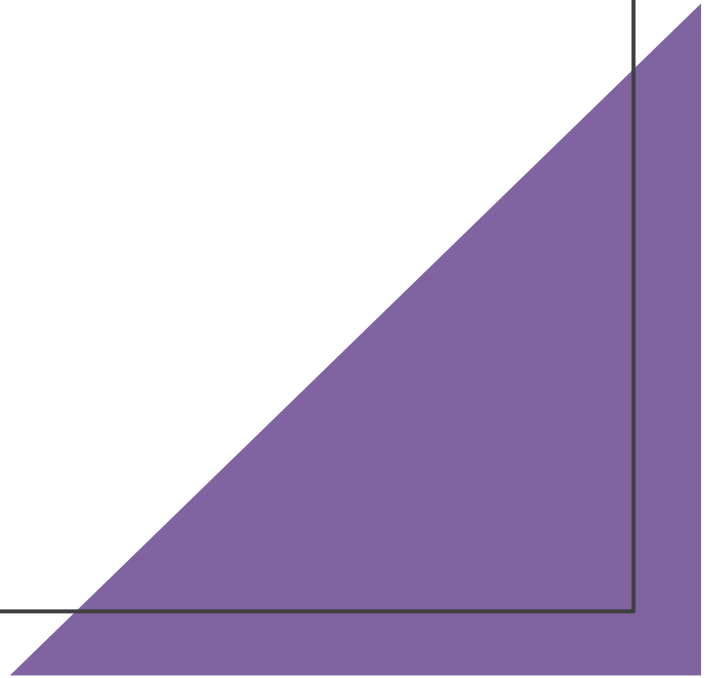
- Inflammatory muscle diseases
- **Polymyositis**
 - Proximal muscle weakness
 - No skin involvement
- **Dermatomyositis**
 - Muscle weakness + characteristic skin rash
- **Key signs:**
 - Gottron papules
 - Heliotrope rash
- **Autoantibodies:**
Anti-Jo-1, anti-Mi-2

POLYMYOSITIS AND DERMATOMYOSITIS

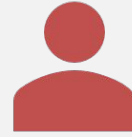


Sjögren's Syndrome

- Autoimmune exocrinopathy
- **Key Features**
 - Dry eyes (xerophthalmia)
 - Dry mouth (xerostomia)
 - Parotid gland enlargement
- **Autoantibodies:**
 - Anti-Ro (SSA), Anti-La (SSB)



Mixed Connective Tissue Disease (MCTD)



Overlap syndrome (**Systemic Lupus Erythematosus (SLE), Systemic Sclerosis (SSc), and Polymyositis (PM).**)



High titers of **anti-U1 RNP antibodies**



Clinical features:

Raynaud
phenomenon
Arthritis
Myositis
Esophageal
dysmotility

Vasculitis

- Inflammation of blood vessel walls
- Can be primary or secondary to CVDs
- Examples:
 - Polyarteritis nodosa
 - ANCA (Anti-Neutrophil Cytoplasmic Antibodies)-associated vasculitis
 - Lupus vasculitis



Common Clinical Manifestations Across CVDs

Fatigue, fever, weight loss

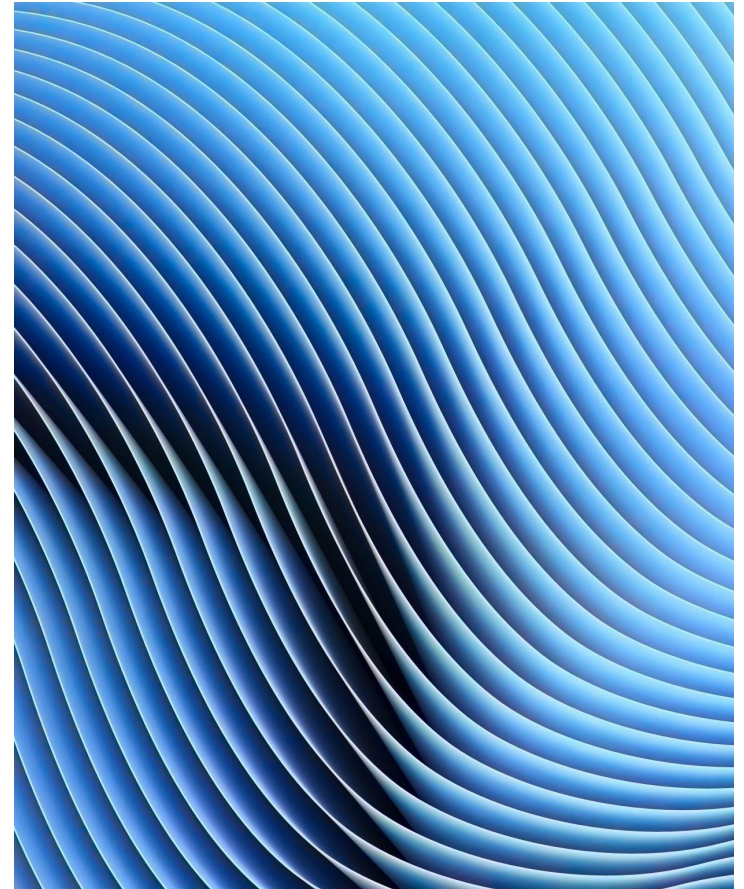
Arthralgia / arthritis

Skin rashes

Raynaud phenomenon

Muscle weakness

Organ-specific involvement



Diagnostic Approach

Laboratory

- ANA (screening)
- Disease-specific autoantibodies
- ESR, CRP
- Complement levels

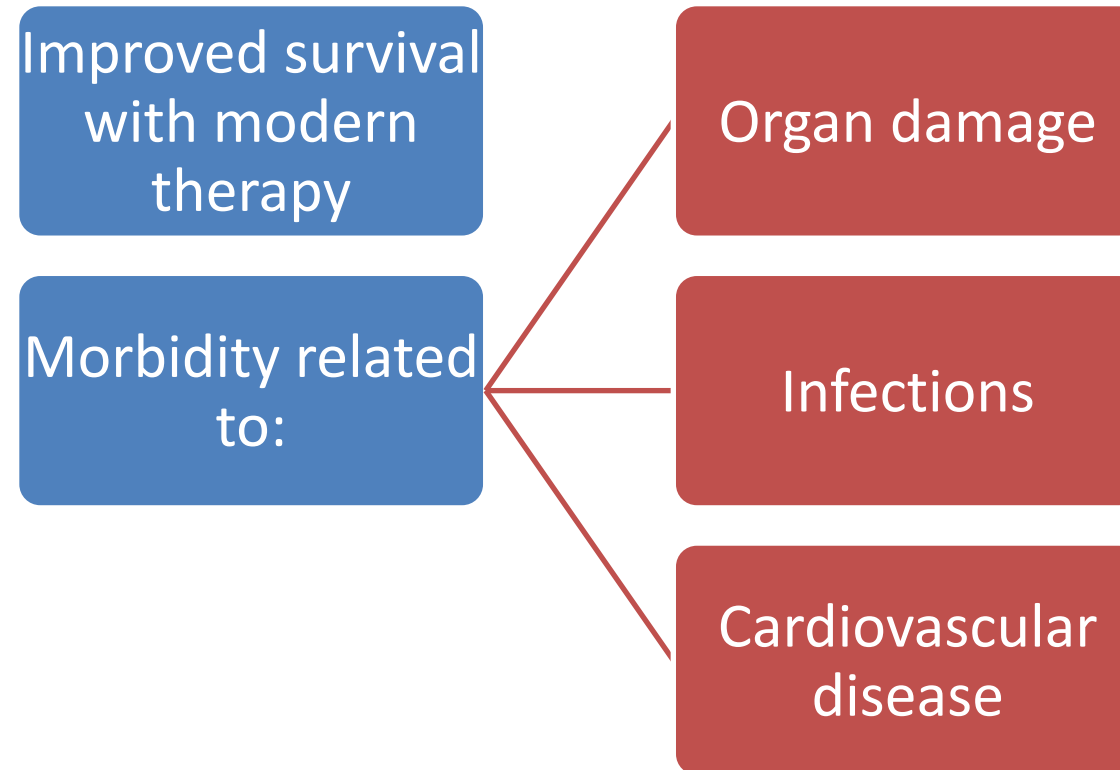
Imaging / Procedures

- X-ray, CT, MRI
- Biopsy (kidney, muscle, skin)

Treatment Principles (General)

- Early diagnosis and treatment
- Immunosuppression tailored to severity
- Minimize long-term corticosteroid use
- Organ-specific therapy
- Multidisciplinary care
- **Drug Classes**
 - NSAIDs
 - Corticosteroids
 - Conventional DMARDs
 - Biologics and targeted therapies

Prognosis



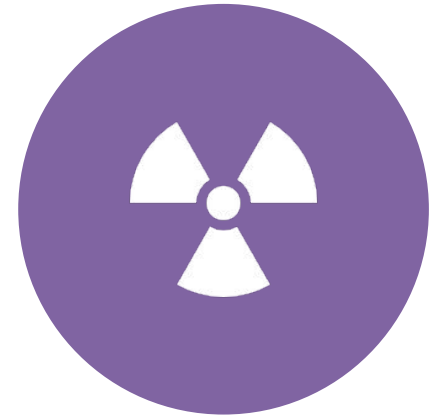
Monitoring



PULMONARY
HYPERTENSION



INTERSTITIAL LUNG DISEASE
(ILD) PROGRESSION



TOXICITY MONITORING

Key Take-Home Messages



Collagen-vascular diseases are systemic autoimmune disorders



Clinical overlap is common



Autoantibodies guide diagnosis



Management is long-term and individualized



Early intervention improves outcomes

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